

THE CALIPER TEST**Tier 2a :: p-Hacking / Selective Reporting**

At a glance: The caliper test detects an excess of statistically significant results that fall just below a reporting threshold — most commonly $p < .05$ — relative to results that fall just above it. A spike near the threshold is a signature of selective stopping, optional stopping rules, or rounding of p-values to achieve significance. The test requires only reported p-values or test statistics from a set of studies; no raw data is needed.

1. Overview

The caliper test was introduced by Alan Gerber and Neil Malhotra in 2008 in the context of empirical sociology, where they observed an implausible discontinuity in the distribution of published z-scores around the conventional significance threshold. The intuition behind the test is a direct extension of the observation that if researchers are free to choose when to stop collecting data, which covariates to include, or how to round reported statistics, they will systematically produce results that land just on the right side of the threshold rather than just above it.

Under honest research practice, the distribution of test statistics near any threshold should be smooth. If a true effect exists, significant and near-significant results should differ only by sampling variation, and results just below .05 should be approximately as common as results just above it. A sharp spike below the threshold — and a corresponding trough just above it — is consistent with only two explanations: strategic manipulation of the analysis process, or systematic rounding of p-values that are close to but above .05 down to just below it.

The caliper test is deliberately simple and conservative. It does not model the full distribution of p-values (as p-curve and z-curve do), and it does not require assumptions about the true underlying effect size (as p-uniform does). It makes only one assumption: that in the absence of manipulation, the count of results just below the threshold should not significantly exceed the count just above it within a narrow symmetric window. This simplicity is both its strength — it is robust to many distributional assumptions — and its limitation, since it discards information from the full distribution.

2. Methodological Overview

The analyst defines a threshold (typically $p = .05$) and a caliper — a symmetric window of fixed width around that threshold. The caliper captures results in two bins: those falling between the threshold and the lower boundary of the caliper window (just significant), and those falling between the threshold and the upper boundary (just non-significant). The test asks whether the count in the just-significant bin significantly exceeds the count in the just-non-significant bin.

Basic binomial test

In the simplest implementation, the counts in the two bins are compared using a one-tailed binomial test. Under the null hypothesis of no manipulation, results are equally likely to fall in either bin, so the expected split is 50/50. An excess of results just below the threshold produces a binomial p-value indicating the probability of observing at least that many results in the lower bin by chance.

`n_below = count of results in (threshold - caliper, threshold]`

e.g., for threshold = .05 and caliper = .01: results with p in (.04, .05]

`n_above = count of results in [threshold, threshold + caliper)`

e.g., for threshold = .05 and caliper = .01: results with p in [.05, .06)

`H0: P(result falls below threshold) = 0.5`

Test statistic: $z = (n_below - n_above) / \sqrt{n_below + n_above}$. Under H_0 , $z \sim N(0,1)$. A significantly positive z indicates excess just-significant results.

Caliper width selection

The caliper width is the most consequential analyst decision. A very narrow caliper (e.g., .005) captures only the most extreme rounding and is highly sensitive to the specific manipulation being detected but has low statistical power with small samples. A wide caliper (e.g., .025) has more power but captures results that

may have arisen for legitimate reasons other than manipulation — a study with a true effect near the threshold will naturally produce results in the window with some frequency.

Gerber and Malhotra (2008) recommended a caliper of .025 (capturing results in the range .025 to .050 and .050 to .075) for their initial application. Subsequent researchers have recommended conducting sensitivity analyses across multiple caliper widths and checking whether the excess is consistent across them — a spike that appears only at one specific caliper width is more likely to be a statistical artifact than a genuine manipulation signal.

Regression discontinuity extension

A more powerful variant, developed by Egger and colleagues and applied extensively in economics, treats the p-value distribution as a continuous function and tests for a discontinuity at the threshold using local polynomial regression. This approach uses more of the distribution's information than the simple bin comparison and allows formal estimation of the size of the discontinuity — the excess density at the threshold — rather than just its statistical significance.

f(p) = α + β1p + β2p^2 + γ · N(p ≤ threshold) + ε

Local polynomial regression on the p-value density. The discontinuity estimate γ measures the excess density at the threshold. A significantly positive γ indicates a spike consistent with selective reporting.

3. Example Application: Sociology and Political Science Journals

Gerber and Malhotra's original 2008 paper applied the caliper test to articles published in the American Journal of Sociology and the American Journal of Political Science between 1995 and 2005. They converted all reported z-scores, t-statistics, and F-statistics to equivalent two-tailed p-values and binned the results within a caliper of .025 around p = .05.

The results were striking: there were approximately 2.5 times as many results in the just-significant bin (.025 to .050) as in the just-non-significant bin (.050 to .075). Across both journals combined, the binomial test was highly significant. The distribution showed a pronounced spike immediately below the threshold, with a corresponding trough immediately above it — consistent with widespread optional stopping or threshold-seeking behavior in both disciplines.

A second analysis in the same paper examined the p = .10 threshold, which some authors treat as marginally significant. A smaller but still substantial spike appeared there as well — consistent with researchers targeting multiple thresholds depending on the strength of their effect and the conventions of their subfield.

Subsequent applications by Head et al. (2015, PLOS Biology) found similar spikes across a sample of over 100,000 p-values extracted from the biomedical and behavioral literature using automated text mining. The pattern was not uniform across fields: experimental fields with pre-registered hypotheses showed smaller spikes than observational fields with more flexible analytic choices. Caliper analysis has since been applied to economics, education research, management, and clinical trial reporting, consistently finding evidence of threshold-targeting behavior, though the magnitude varies substantially by field and publication venue.

4. Requirements & Applicability

The caliper test's principal advantage is its minimal data requirements. It operates on the marginal distribution of reported p-values, which are available from any study that reports hypothesis tests.

Reported p-values or test statistics	The test requires the exact reported p-value for each hypothesis test, or a test statistic (z, t, F, chi-squared) and degrees of freedom from which an exact p-value can be recovered. Approximate p-values (p < .05, p = NS) cannot be used — they provide no information about whether the result fell just above or just below the threshold.
Minimum study count	The binomial test is unreliable with fewer than 20–30 results in the combined window. A caliper of .025 on each side captures a narrow slice of the distribution; many studies produce no results in this window. Practical application typically requires at least 50–100 papers to generate enough observations in the caliper window for meaningful inference.

Consistent threshold	The test is defined relative to a specific threshold. Most applications use $p = .05$, but the test can be applied to any threshold where selective reporting is plausible ($p = .10$, $p = .01$). Analyzing multiple thresholds simultaneously requires multiple comparison correction.
One test per study or all tests?	A consequential analyst choice: using only the focal hypothesis test per study reduces false positives but reduces power; using all reported tests inflates the sample but may mix primary and exploratory results. The choice should be documented and sensitivity analyses conducted across both approaches.
Raw data required?	No. The test operates on reported p-values in published papers. Automated extraction tools (statcheck, custom parsers) can extract p-values from PDF text at scale, making the caliper test one of the more computationally scalable detection methods.
Directional hypotheses	Results reported as one-tailed tests should be treated carefully. A one-tailed p of .049 is equivalent in magnitude to a two-tailed p of .098 — combining one- and two-tailed tests without conversion will distort the distribution near the threshold.

5. Limitations

- The caliper test is sensitive to the choice of caliper width. Different widths can produce qualitatively different conclusions from the same dataset. Reporting sensitivity analyses across widths (e.g., .010, .025, .050) is essential but not always done.
- The test cannot distinguish rounding from strategic stopping. A researcher who rounds $p = .052$ to $p = .05$ in a table produces the same distributional signal as a researcher who added data points until p crossed .05. The test detects the signature of threshold-seeking but cannot identify which mechanism produced it.
- Legitimate effects near the threshold produce false signals. If a literature contains many studies of a true effect whose power is near 50%, approximately half of results will fall just below the threshold and half just above — not because of manipulation, but because the effect is real and borderline. Caliper tests applied to such a literature will flag excess just-significant results that are entirely legitimate.
- The test does not account for publication bias in the broader sense. Even without any within-study manipulation, if journals preferentially publish significant results, the caliper window will be depopulated above the threshold (null results withheld) and populated below it (significant results published). The spike signature is produced by selection at the journal level, not necessarily the researcher level. Caliper analysis cannot distinguish these two sources.
- P-value reporting conventions vary across fields and decades. Some journals report p-values to two decimal places (rounding .049 to .05), some to three. This rounding creates artificial spikes at round-number p-values that are not evidence of manipulation. Analyses should check for and account for discretization artifacts.
- The test has low power in small literature samples. With fewer than 50 studies, the caliper window will typically contain too few observations for stable inference. Fields with small literatures — specialized subfields, emerging research areas — are poorly served by this method.

6. How to Apply

Step 1 — Collect and convert p-values

Identify the set of studies (defined by inclusion criteria specified before examining results). For each study, record the focal hypothesis test statistic and convert to an exact two-tailed p-value. If the paper reports only an inequality ($p < .05$), exclude that result from the analysis — it cannot be placed precisely in the distribution. Automated extraction using statcheck or custom PDF parsers can scale this step to hundreds of papers.

Step 2 — Choose threshold and caliper width

Select the threshold of interest (typically $p = .05$) and the caliper width. For an initial analysis, a width of .025 on each side is a reasonable default, capturing results in the ranges (.025, .050] and [.050, .075). Document the choice and plan sensitivity analyses at widths of .010 and .050.

Step 3 — Count results in each bin

Count the number of results falling in the just-significant bin (below the threshold within the caliper) and the just-non-significant bin (above the threshold within the caliper). If analyzing multiple thresholds, repeat for each.

Step 4 — Run the binomial test

Compute the binomial test statistic: $z = (n_{\text{below}} - n_{\text{above}}) / \sqrt{n_{\text{below}} + n_{\text{above}}}$. Compare to the standard normal distribution for a one-tailed test (H_1 : excess below threshold). In R, this is: `binom.test(n_below, n_below + n_above, p = 0.5, alternative = 'greater')`.

Step 5 — Visualize the distribution

Plot a histogram of all p-values in the range [0, 1] with bin widths equal to the caliper (e.g., 0.025). Visual inspection of the spike at the threshold is often more informative than the test statistic alone — it reveals whether the excess is localized at the threshold or part of a broader pattern. The `rddensity` package in R implements the regression discontinuity variant with confidence intervals and plotting functions.

Step 6 — Conduct sensitivity analyses

Re-run the analysis at alternative caliper widths (.010, .050). Test the $p = .10$ threshold to check for secondary threshold-seeking. Compare results using all reported tests versus focal tests only. If conclusions are consistent across these variations, confidence in the finding increases substantially.

7. Caliper Test vs. Related Distribution-Based Tools

The caliper test occupies a specific niche among p-value distribution methods. Understanding where it sits relative to p-curve and z-curve helps clarify when to prefer it.

Caliper test	Tests for a spike at a specific threshold using only results in a narrow window. No distributional assumptions beyond smoothness of the underlying density. Very simple to compute and explain. Low power but highly interpretable. Best for documenting threshold-seeking behavior as a descriptive fact about a literature.
p-curve	Uses the full distribution of significant p-values (0 to .05) to diagnose evidential value. More powerful than the caliper test for detecting the presence or absence of a true effect. Cannot localize the manipulation to the threshold region — a left-skewed p-curve could reflect either p-hacking or a very small true effect.
z-curve 2.0	Uses both significant and non-significant test statistics to estimate replication and discovery rates. Provides field-level summaries not available from the caliper test. More complex, requires more studies, but provides richer output. Complementary: use z-curve for field-level diagnosis, caliper test for threshold-specific documentation.
Egger / funnel plot	Detects publication bias across a meta-analytic set via asymmetry in effect size vs. precision. Operates on effect sizes rather than p-values, making it complementary to caliper analysis. A literature can show caliper-test excess (within-study manipulation) without funnel asymmetry (between-study publication bias), or vice versa.

8. Related Tools

p-curve	Examines the shape of the full significant p-value distribution. Where the caliper test asks 'are there too many results just below the threshold?', p-curve asks 'is the shape of the significant distribution consistent with a true effect?' Run both: caliper test for threshold-specific diagnosis, p-curve for evidential value.
statcheck	Automated extraction and recalculation of p-values from paper PDFs. The primary tool for building the p-value dataset required by the caliper test at scale. <code>statcheck</code>

	also flags decision-changing reporting errors that are distinct from (but correlated with) the threshold-seeking behavior the caliper test detects.
z-curve 2.0	Fits a mixture model to all reported z-scores to estimate expected replication and discovery rates. Uses the full distribution rather than just the threshold window, providing richer output. The two tools are best understood as complementary rather than competing.
Excess Significance Test (TES)	Tests whether the observed count of significant results exceeds what would be expected given study power. Where the caliper test examines the distribution near a threshold, TES examines the overall rate of significance across the literature. Convergence of a positive caliper test and a positive TES provides substantially stronger evidence of systematic reporting problems.
Registered Reports	The prospective remedy: publishing a commitment to report all results regardless of significance removes threshold-seeking incentives at the source. Scheel et al. (2021) showed that Registered Reports produce far fewer results just below $p = .05$ than standard publications — the caliper-test distribution is flatter, as expected from genuine rather than threshold-seeking research.

Key Citations

- Gerber, A. S. & Malhotra, N. (2008). Publication bias in empirical sociological research: Do arbitrary significance thresholds distort published results? *Sociological Methods & Research*, 37(1), 3–30. <https://doi.org/10.1177/0049124108318973>
- Gerber, A. S. & Malhotra, N. (2008). Do statistical reporting standards affect what is published? Publication bias in two leading political science journals. *Quarterly Journal of Political Science*, 3(3), 313–326. <https://doi.org/10.1561/100.00008024>
- Head, M. L., Holman, L., Lanfear, R., Kahn, A. T., & Jennions, M. D. (2015). The extent and consequences of p-hacking in science. *PLOS Biology*, 13(3), e1002106. <https://doi.org/10.1371/journal.pbio.1002106>
- Masicampo, E. J. & Lalande, D. R. (2012). A peculiar prevalence of p values just below .05. *Quarterly Journal of Experimental Psychology*, 65(11), 2271–2279. <https://doi.org/10.1080/17470218.2012.711335>
- Brodeur, A., Cook, N., & Heyes, A. (2020). Methods matter: p-hacking and publication bias in causal analysis in economics. *American Economic Review*, 110(11), 3634–3660. <https://doi.org/10.1257/aer.20190687>
- Scheel, A. M., Schijen, M. R. M. J., & Lakens, D. (2021). An excess of positive results: Comparing the standard psychology literature with Registered Reports. *Advances in Methods and Practices in Psychological Science*, 4(2). <https://doi.org/10.1177/25152459211007467>
- Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). P-curve: A key to the file drawer. *Journal of Experimental Psychology: General*, 143(2), 534–547. <https://doi.org/10.1037/a0033242>
- Nuijten, M. B., Hartgerink, C. H. J., van Assen, M. A. L. M., Epskamp, S., & Wicherts, J. M. (2016). The prevalence of statistical reporting errors in psychology (1985–2013). *Behavior Research Methods*, 48, 1205–1226. <https://doi.org/10.3758/s13428-015-0664-2>